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CERAMIC SUSCEPTOR

Abstract

Disclosed is a ceramic susceptor. The ceramic susceptor may include: an insulating plate in which one or more electrodes are arranged; a hollow shaft with one end connected to the insulating plate; and one or more electrode rods connected to the electrodes. The hollow shaft may include one or more side wall holes penetrating an interior of a side wall, and the one or more electrode rods may include one or more first electrode rods, each of which is electrically connected to a first electrode among the one or more electrodes. The first electrode rods may extend through the side wall holes, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based on and claims priority under 35 U.S.C. 119 to Korean Patent Application No. 10-2024-0031493, filed on Mar. 5, 2024, in the Korean Intellectual Property Office, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the invention

[0002] The present disclosure relates to a ceramic susceptor, and more particularly, to a ceramic susceptor in which an electrode rod is arranged within the side wall of a hollow shaft.

2. Description of the Prior Art

[0003] In general, a semiconductor device or a display device is manufactured by sequentially laminating multiple thin film layers including a dielectric layer and a metal layer on a glass substrate, a flexible substrate, or a semiconductor wafer substrate and then patterning the thin film layers. These thin film layers are sequentially deposited on the substrate through a chemical vapor deposition (CVD) process or a physical vapor deposition (PVD) process. The CVD process includes a low-pressure CVD (LPCVD) process, a plasma-enhanced CVD (PECVD) process, a metal-organic CVD (MOCVD) process, and the like.

[0004] Such CVD and PVD apparatuses are equipped with a ceramic susceptor for supporting glass substrates, flexible substrates, semiconductor wafer substrates, and the like, and for heat treatment or the like. The ceramic susceptor may be installed in the CVD and PVD apparatuses to be used for substrate heating during processes such as heat treatment. In addition, the ceramic susceptor may include a radio frequency (RF) electrode and may also be used to form plasma in processes such as an etching process of thin film layers formed on a semiconductor wafer substrate.

[0005] FIG. 1 is a schematic cross-sectional view of a conventional ceramic susceptor 1.

[0006] As illustrated in FIG. 1, a conventional ceramic susceptor 1 includes an insulating plate 10 coupled to a shaft 20. The insulating plate 10 includes electrodes 12 or heating elements 14 arranged between ceramic materials. The shaft 20 is of a hollow type, providing a through-hole that allows for rods 21, 22, and 23 to pass therethrough. The rods are respectively connected to the electrodes 12 or heating elements 14 to supply power.

[0007] However, in the structure of the conventional ceramic susceptor 1 as illustrated in FIG. 1, since the power supply rods 21, 22, and 23 are arranged within the through-hole of the hollow shaft 20, the rods 21, 22, and 23 are prone to move or shake within the relatively large through-hole. This may cause the rods 21, 22, and 23 to physically bend or short-circuit.

[0008] In addition, in order to prevent arcing caused by electrical contact, it is necessary to insulate the rods 21, 22, and 23 from each other within the same space in the through-hole of the shaft 20. To this end, the rods 21, 22, and 23 are wrapped and insulated with separate insulators made of a material such as Al.sub.2O.sub.3. However, since the insulators are coated in the form of a pipe, the spacing between the rods becomes narrower, and if not coated, arcing may be caused.

[0009] In addition, from a design perspective, the metal rods 21, 22, and 23 are densely packed in the center of the through-hole in the shaft 20. Therefore, when it is necessary to change the number or positions of the rods depending on the situation, there are limitations in changing the number or positions of the rods due to the spatial constraints within the through-hole of the shaft 20.

SUMMARY OF THE INVENTION

[0010] Accordingly, the present disclosure has been made to solve the aforementioned problems, and provides a ceramic susceptor in which electrode rods are arranged in side wall holes (e.g., $\Phi 3$ to $\Phi 6$) of a hollow shaft to eliminate the space in which the electrodes may move or bend, thereby making it easier to prevent the risk of physical short circuits.

[0011] In addition, the present disclosure provides a ceramic susceptor in which electrode rods may be arranged within the hollow of a hollow shaft, but arrangement of more electrode rods is possible without arranging electrode rods within the hollow, and in which free design of electrode arrangement is possible depending on changes in the number or positions of electrode rods.

[0012] Furthermore, the present disclosure provides a ceramic susceptor in which by arranging electrode rods in side wall holes (e.g., $\Phi 3$ to $\Phi 6$) of a hollow shaft, the insulation of a hollow shaft is utilized so that a separate insulator coating for insulating each electrode rod is unnecessary, and in which the radio frequency electrode rods and electrode rods for a heater are separately arranged in the side wall holes of the hollow shaft or are divisionally arranged in the side wall holes and the hollow shaft, ensuring that the gap between the rods does not become narrow, thereby preventing the occurrence of arcing.

[0013] In summary, in view of the foregoing, a ceramic susceptor according to an aspect of the present disclosure may include: an insulating plate in which one or more electrodes are arranged; a hollow shaft with one end connected to the insulating plate; and one or more electrode rods connected to the electrodes. The hollow shaft may include one or more side wall holes penetrating an interior of a side wall, and the one or more electrode rods may include one or more first electrode rods, each of which is electrically connected to a first electrode among the one or more electrodes. The first electrode rods may extend through the side wall holes, respectively.

[0014] The one or more electrode rods further include one or more second electrode rods, each of which is connected to a second electrode among the one or more electrodes. The one or more second electrode rods may extend through a hollow of the hollow shaft.

[0015] The first electrodes may each include a heater electrode, and the second electrodes may each include a radio frequency electrode. At this time, multiple first electrode rods may be respectively connected to terminals of the heater electrodes.

[0016] The first electrodes may each include a radio frequency electrode, and the second electrodes may each include a heater electrode. At this time, multiple first electrode rods may be respectively connected to terminals of the heater electrodes.

[0017] The electrode rods may each include a gas flow path, and one or more gas holes in communication with the gas flow path.

[0018] The ceramic susceptor may further include a conductive socket coupled to the electrode rods and configured to supply power, and the conductive socket may include a gas supply flow path configured to communicate with the gas flow path.

[0019] The ceramic susceptor may further include a mount coupled to the lower portion of the hollow shaft, and a sealing member coupled between the hollow shaft and the mount.

[0020] The hollow shaft may include a groove formed on a surface coupled to the mount and extending to meet the side wall holes at least once, the mount may include a gas supply flow path configured to communicate with the groove, and the one or more first electrode rods may pass through the groove.

[0021] The one or more electrode rods may further include one or more second electrode rods, each of which is connected to a second electrode among the one or more electrodes, the one or more electrode rods extending through a hollow of the hollow shaft, and the one or more second electrode rods may pass through the mount.

[0022] The sealing member may be a ring structure, the ring-shaped sealing member may include a rectangular, circular, oval, or trapezoidal cross-sectional shape, and the sealing member may be made of a silicone material.

[0023] According to the ceramic susceptor of the present disclosure, by arranging the electrode rods in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) of the hollow shaft, the space in which the electrodes shake or bend is eliminated, which makes it easy to prevent the risk of physical short circuits.

[0024] In addition, the arrangement of the electrode rods inside the hollow of the hollow shaft may be combined with the arrangement of the electrode rods in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) of

the hollow shaft, allowing for flexible design of electrode arrangement depending on changes in the number or positions of the electrode rods, and enabling the arrangement of more electrode rods. [0025] Furthermore, by arranging electrode rods in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) of the hollow shaft, the insulation of the hollow shaft is utilized so that a separate insulator coating for insulating each electrode rod is unnecessary. In addition, the radio frequency electrode rods and electrode rods for a heater are separately arranged in the side wall holes of the hollow shaft or are divisionally arranged in the side wall holes and the hollow shaft, ensuring that the gap between the rods does not become narrow, thereby preventing the occurrence of arcing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The above and other aspects, features and advantages of the present disclosure will be more apparent from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0027] FIG. 1 is a schematic cross-sectional view of a conventional ceramic susceptor;

[0028] FIG. 2 is a schematic cross-sectional view of a ceramic susceptor according to an embodiment of the present disclosure;

[0029] FIGS. 3A to 3D are cross-sectional views taken along line A-B in FIG. 2, illustrating embodiments in each of which multiple electrode rods are respectively arranged in side wall holes of the shaft;

[0030] FIG. 4A is a cross-sectional view illustrating a ceramic susceptor according to another embodiment of the present disclosure;

[0031] FIG. 4B is an enlarged view of a power supply conductor socket in FIG. 4A;

[0032] FIG. 4C is an enlarged view of an electrode rod having a gas flow path and gas hole(s) in FIG. 4A;

[0033] FIG. 5A is a cross-sectional view illustrating a ceramic susceptor according to yet another embodiment of the present disclosure; and

[0034] FIG. 5B is a view of a surface C-D of the hollow shaft to which the mount is coupled, as seen from below in FIG. 5A.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0035] Hereinafter, the present disclosure will be described in detail with reference to the accompanying drawings. Herein, like components in each drawing are denoted by like reference numerals if possible. In addition, detailed descriptions of already known functions and/or configurations will be omitted. In the following description, components necessary for understanding operations according to various embodiments will be mainly described, and descriptions of elements that may obscure the gist of the description will be omitted. In addition, some elements in the drawings may be exaggerated, omitted, or schematically illustrated. The size of each component does not entirely reflect the actual size. Therefore, the descriptions provided herein are not limited by the relative sizes or spacings of the components drawn in each drawing.

[0036] In describing the embodiments of the present disclosure, when a detailed description of the known technology related to the present disclosure is determined to unnecessarily obscure the subject matter of the present disclosure, the detailed description will be omitted. In addition, terms to be described later are defined in consideration of functions in the present disclosure, and may vary according to the intention, custom, or the like of a user or operator. Therefore, the definitions of the terms should be made based on the description throughout this specification. Terms used in the detailed description are only for describing the embodiments of the present disclosure, and should not be construed as limiting in any way. Unless expressly used otherwise, singular expressions include the meanings of plural expressions. As used herein, expressions such as

“including” or “comprising” are intended to indicate any features, numbers, steps, operations, elements, or some or combinations thereof, and should not be construed to exclude the existence or possibility of one or more other features, numbers, steps, operations, elements, or some or combinations thereof, in addition to those described above.

[0037] In addition, terms such as “first” and “second” may be used to describe various components, but the components are not limited by the terms, and these terms are only used for the purpose of distinguishing one component from another.

[0038] First, as described below, a ceramic susceptor **100**, **200**, or **300** of the present disclosure may be installed in a chamber of a CVD apparatus or a PVD apparatus to be used for heating using heating elements (electrodes **114**) and/or supporting a substrate **11** using a (radio frequency) electrode **112**, or in processes such as plasma-enhanced chemical vapor deposition or dry etching using plasma. The ceramic susceptor **100**, **200**, or **300** of the present disclosure may further include a chuck electrode (not illustrated) arranged at a predetermined distance from the electrode **112** to enable chucking and de-chucking of the substrate **11**. In some cases, the electrode **112** may be a radio frequency electrode arranged to receive power for plasma generation as described above, but may also function as an electrostatic chuck electrode (or chuck electrode) arranged to receive power for chucking and de-chucking the substrate **11**. The substrate **11** may be a processing target substrate for various applications, such as a semiconductor wafer, a glass substrate, or a flexible substrate.

[0039] Accordingly, the insulating plate **110** may be configured to stably support a processing target substrate while enabling various semiconductor processes such as heating using the heating element **114** and/or plasma-enhanced chemical vapor deposition or plasma dry etching using the electrode **112**.

[0040] FIG. **2** is a schematic cross-sectional view of a (ceramic) susceptor **100** according to an embodiment of the present disclosure.

[0041] Referring to FIG. **2**, a (ceramic) susceptor **100** according to an embodiment of the present disclosure may include an insulating plate **110** made of a ceramic material, a hollow shaft **120** having a hollow **129**, and may further include a mount **140** as needed. The insulating plate **110**, the shaft **120**, and the mount **140** may be sequentially connected, and a sealing member **144** with a ring structure may be provided between the shaft **120** and the mount **140**. The sealing member **144** may be configured to seal the surfaces where the lower portion of the hollow shaft **120** and the mount **140** are coupled.

[0042] According to an embodiment of the present disclosure, the ceramic susceptor **100** has electrode rods **121** arranged in side wall holes **125** (e.g., $\Phi 3$ to $\Phi 6$) of the hollow shaft **120**, the spaces in which the electrode rods **121** may move or bend are eliminated, thereby facilitating prevention of the risk of physical short circuits. In addition, an electrode rod **122**, penetrating through the hollow **129** of the hollow shaft **120**, may further be arranged. The electrode rods **121** may be connected to the electrodes **112** or **114**, and if needed, the electrode rod **122** may also be connected to the electrodes **112** or **114**.

[0043] To this end, the insulating plate **110** may be configured such that one or more heating elements or electrodes **114** are arranged (embedded) between the ceramic materials, and in some cases, the insulating plate **110** may be further configured such that one or more other electrodes **112** are arranged (embedded) at a predetermined distance from the heating elements **114**. The insulating plate **110** is configured to stably support a processing target substrate **11** while enabling various semiconductor processes, such as heating using the heating element **114** and/or supporting the substrate, plasma-enhanced chemical vapor deposition, or plasma dry etching using the electrode **112**. The insulating plate **110** may be a plate-like structure having a predetermined shape. For example, the insulating plate **110** may be a circular plate-like structure, but is not necessarily limited thereto. Here, the ceramic material may be at least one of Al.sub.2O.sub.3, Y.sub.2O.sub.3, Al.sub.2O.sub.3/Y.sub.2O.sub.3, ZrO.sub.2, autoclaved lightweight concrete (ALC), TiN, AlN, TiC,

MgO, CaO, CeO.sub.2, TiO.sub.2, B.sub.xC.sub.y, BN, SiO.sub.2, SiC, YAG, mullite, and AlF.sub.3, preferably aluminum nitride (AlN). Furthermore, the powder of the ceramic material may be molded and sintered to provide the insulating plate **110**, and each ceramic powder for this purpose may optionally contain about 0.1 to 10%, preferably about 1 to 5% of yttrium oxide powder.

[0044] In addition, the heating element **114** may be formed in a planar coil shape or a flat plate shape using a heating wire (or resistance wire). In addition, the heating element **114** may be fabricated in a multi-layer structure for precise temperature control. The heating element **114** may be connected to a power source for a separate heating element **114** via connection rods **121** and **122** for power supply. The heating element may perform the function of heating a processing target substrate **11** on the insulating plate **110** to a predetermined temperature for substrate heating, vapor deposition, etching, and the like in a semiconductor process. The connection rods **121** and **122** may pass through the hollow **129** of the shaft **120** or the side wall through-holes (side wall holes) **125**, and may extend to the outside through the mount **140**.

[0045] The electrode **112** (including a chuck electrode) may be made of tungsten (W), molybdenum (Mo), silver (Ag), gold (Au), niobium (Nb), titanium (Ti), aluminum nitride (AlN), or an alloy thereof, preferably molybdenum (Mo). The electrode **112** may be connected to a power terminal (e.g., a ground) via another connection rod (not illustrated). The connection rods **121** and **122** may pass through the hollow **129** of the shaft **120** or the side wall through-holes (side wall holes) **125**, and may extend to the outside through the mount **140**. For example, the electrode **112** may be used as a chuck electrode for an electrostatic chuck function to support a substrate **11** placed on the insulating plate **110** or as a plasma-generating electrode for plasma generation functions in processes such as plasma-enhanced chemical vapor deposition or dry etching in reactive ion etch (RIE) equipment.

[0046] The connection rods **121** and **122** may be bonded to electrode pads **51** through brazing. The electrode pads **51** may also be joined to the electrodes **112** and **114** of the insulating plate **110** through brazing. For example, for the brazing, each joint may be formed by injecting a conductive filler, heating at a high temperature, and cooling.

[0047] The shaft **120** has a pipe shape with a hollow **129**, and one end is joined to the bottom surface of the insulating plate **110**. The shaft **120** may be made of the same ceramic material as the insulating plate **110** and coupled to the same. Here, the ceramic material may be at least one of Al.sub.2O.sub.3, Y.sub.2O.sub.3, Al.sub.2O.sub.3/Y.sub.2O.sub.3, ZrO.sub.2, autoclaved lightweight concrete (ALC), TiN, AlN, TiC, MgO, CaO, CeO.sub.2, TiO.sub.2, B.sub.xC.sub.y, BN, SiO.sub.2, SiC, YAG, mullite, and AlF.sub.3, preferably aluminum nitride (AlN). Furthermore, the powder of the ceramic material may be molded and sintered to provide the shaft **120**, and each ceramic powder for this purpose may optionally contain about 0.1 to 10%, preferably about 1 to 5% of yttrium oxide powder. The shaft **120** may be joined to the insulating plate **110** using a bonding material such as ceramic paste. In some cases, the shaft **120** may be mechanically coupled to the insulating plate **110** by using bolts, nuts, and the like.

[0048] FIGS. 3A to 3D are cross-sectional views taken along line A-B in FIG. 2, illustrating embodiments in each of which multiple electrode rods **121** are arranged in side wall holes **125** of the shaft **120**.

[0049] Referring to FIGS. 3A to 3D, the shaft **120** may include one or more side wall holes **125** penetrating the side wall, and electrode rods **121** may be respectively arranged in the side wall holes **125** of the shaft **120**. That is, one or more electrode rods **121** are connected to one or more electrodes **112** or **114** and arranged to extend through respective side wall holes **125**. In addition, one or more electrode rods **122** may be connected to the one or more electrodes **112** or **114** and arranged to extend through the hollow **129** from the opposite side of the shaft **120**. Any one of the electrode rods **121** and **122** may be omitted. In other words, only the electrode rods **121** may be arranged in the side wall holes **125**.

[0050] Accordingly, each connection rod **121** or **122** extends through the hollow **129** of the shaft **120** or through each side wall hole **125**. Furthermore, when a mount **140** is provided, each connection rod **121** or **122** may extend to the outside further through the mount **140**.

[0051] For example, preferably, among the one or more electrodes **112** and **114**, a first electrode **112** may be a radio frequency electrode as described above, and a second electrode **114** may be a heater electrode, such as the above-described heating element. Conversely, among the one or more electrodes **112** and **114**, the first electrode **112** may be a heater electrode, such as the above-described heating element, and the second electrode **114** may be a radio frequency electrode as described above.

[0052] For example, FIG. **3A** may illustrate two electrode rods **121** arranged for a single-zone heating element (heater electrode). Specifically, a pair of heating elements (heater electrodes) **121** may preferably be connected to opposite terminals (not illustrated) of the second electrode **114**, which serves as a heater electrode. This corresponds to a case where the heating element (pattern) of the second electrode **114**, which is a heater electrode, is electrically connected between the two opposite terminals (not illustrated) without interruption, thereby forming a single zone and generating heat. At this time, one or more first electrodes **112**, which are radio frequency electrodes, may be connected to one or more electrode rods **122**, respectively.

[0053] In addition, as described above, the functions of the first electrode **112** and the second electrode **114** may be interchanged. The configuration, where the electrode rods **121** and **122** are interchanged for opposite functions, may be applied as needed. Furthermore, the inner surface of the side wall of the shaft **120** may be a plain form with no protrusions, as illustrated in FIGS. **3A** and **3C**, or it may have an uneven form with protrusions, as illustrated in FIGS. **3B** and **3D**. In the case of FIGS. **3B** and **3D**, one side wall hole **125** for an electrode rod **121** or **122** may be arranged in each of radially inward convex portions. Such an uneven form of the inner surface of the side wall of the shaft **120** may make it easy to distinguish the positions where the electrode rods **121** and **122** are arranged and is advantageous in ensuring the internal space of the hollow **129** of the shaft **120**, thereby allowing for more flexible arrangement of structures, such as the rods **122** placed inside.

[0054] In addition, FIG. **3B** may illustrate four electrode rods **121** for the arrangement of two-zone heating elements (heater electrodes). That is, two of the four electrode rods **121** may be connected to a first pair of terminals (not illustrated) of the second electrode **114** as the heater electrode, and a heating element electrode (pattern) electrically extending without interruption between the first pair of terminals (not illustrated) may form a first zone. The remaining two of the four electrode rods **121** may be connected to a second pair of terminals (not illustrated) of the second electrode **114** as the heater electrode, and a heating element electrode (pattern) electrically extending without interruption between the second pair of terminals (not illustrated) may form a second zone. At this time, one or more first electrodes **112**, which are radio frequency electrodes, may be connected to one or more electrode rods **122**, respectively.

[0055] In a manner similar to that illustrated in FIG. **3A** or FIG. **3B**, as illustrated in FIGS. **3C** and **3D**, eight electrode rods **121**, grouped in pairs, may be connected to the respective terminal pairs of four heating element electrodes (patterns) included in the second electrode **114** as the heater electrode, thereby forming four heating zones. Here, one or more first electrodes **112**, which are radio frequency electrodes, may also be connected to one or more electrode rods **122**, respectively.

[0056] FIG. **4A** is a cross-sectional view illustrating a ceramic susceptor **200** according to another embodiment of the present disclosure. FIG. **4B** is an enlarged view of a power supply conductor socket **150** in FIG. **4A**.

[0057] FIG. **4C** is an enlarged view of an electrode rod **123** having a gas flow path **123-1** and gas holes **123-2** in FIG. **4A**.

[0058] Referring to FIGS. **4A** to **4C**, the ceramic susceptor **200** is a modification of the ceramic susceptor **100** of FIG. **2**. The ceramic susceptor **200** includes a power supply conductor socket **150**

and an electrode rod **123** arranged within a side wall hole **125** and coupled to the power supply conductor socket. The ceramic susceptor **200** may include the power supply conductor socket **150** and the electrode rod **123** in addition to the ceramic susceptor **100** of FIG. **2**. In some cases, the ceramic susceptor **200** may include the power supply conductor socket **150** and the electrode rod **123** in place of the electrode rod **121** in the side wall hole **125** of the ceramic susceptor **100** of FIG. **2**.

[0059] First, in FIG. **4A**, each electrode rod **121** or **122**, which supplies power to the electrode **112** or **114**, may be accommodated in the hollow **129** or a side wall hole **125** of the shaft **120**, may extend to the outside through the mount **140**.

[0060] In addition, as illustrated in FIG. **4A**, the mount **140** is connected to the longitudinal end of the shaft **120**. The mount **140** may be mechanically coupled to the longitudinal end of the shaft **120**. The connection between the shaft **120** and the mount **140** may be made by mechanical coupling using bolts, nuts, or the like. At this time, a sealing member **144** may be provided between the lower portion of the hollow shaft **120** and the mount **140** for sealing. The sealing member **144** may be a ring-shaped member for sealing the surfaces where the lower portion of the hollow shaft **120** and the mount **140** are coupled, and its cross-sectional shape may take various forms, such as rectangular, trapezoidal, circular, or elliptical. Preferably, the sealing member **144** is made of silicone material. For example, a predetermined groove may be provided on the upper portion of the mount **140** in advance radially outward from the side wall hole **125** to accommodate the sealing member **144**. The sealing member **144** may be placed higher than the predetermined groove and then compressed by the aforementioned mechanical coupling, thereby sealing the surfaces where the lower portion of the shaft **120** and the mount **140** are coupled.

[0061] The aforementioned connection mount **140** may be made of a ceramic material as described above. In some cases, the mount **140** may be made of a metallic material such as aluminum (Al), and in such cases, it is desirable to insulate the necessary portions through which the connection rods **121** or **122** pass. The ceramic material may be at least one of Al.sub.2O.sub.3, Y.sub.2O.sub.3, Al.sub.2O.sub.3/Y.sub.2O.sub.3, ZrO.sub.2, autoclaved lightweight concrete (ALC), TiN, AlN, TiC, MgO, CaO, CeO.sub.2, TiO.sub.2, B.sub.xC.sub.y, BN, SiO.sub.2, SiC, YAG, mullite, and AlF.sub.3, preferably aluminum nitride (AlN). Furthermore, the powder of the ceramic material may be molded and sintered to provide the shaft **120**, and each ceramic powder for this purpose may optionally contain about 0.1 to 10 wt. %, preferably about 1 to 5 wt. %, of yttrium oxide powder.

[0062] In FIG. **4B**, the conductor socket **150** may be made of a conductor such as gold, silver, copper, or nickel, and may include a body **151**, and a gas connection portion **152**, a power connection portion **153**, and a rod connection portion **154**, which are either coupled to or integrated with the body **151**.

[0063] As illustrated in FIG. **4C**, the electrode rod **123** may have a form in which a gas flow path **123-1** is provided longitudinally as an empty space inside the electrode rod, and one or more gas holes **123-2** may be formed on the side surface of the electrode rod to communicate with the gas flow path **123-1**.

[0064] The electrode rod **123** is coupled to the rod connection portion **154** of the conductor socket **150**, and the gas flow path **123-1** may be provided from the end of the rod connection portion **154** side of the electrode rod **123** by a predetermined length toward the insulating plate **110** side or to the end on the insulating plate **110** side. Multiple gas holes **123-2** may be provided in a straight line along the side surface of the electrode rod **123** at predetermined intervals, but in some cases, they may not be provided in a straight line and may be provided at different intervals.

[0065] The conductor socket **150** is configured to supply power for a heater, power for a radio frequency electrode, or power for a chuck electrode from a power line connected to the power connection portion **153** to the electrode rod **123** coupled to the rod connection portion **154**. In addition, the conductor socket may also provide an inert gas (e.g., He or N.sub.2) to the electrode

rod **123** through the gas supply flow path **155** between the gas connection portion **152** and the rod connection portion **154**.

[0066] A gas tube for providing an inert gas (e.g., He or N.sub.2) may be coupled to the gas connection portion **152**, and the gas supplied to the electrode rod **123** through the gas supply flow path **155**, which communicates with the gas tube, may flow along the gas flow path **123-1**, which communicates with the gas supply flow path **155**, and may be supplied to the side wall hole **125** through the gas hole(s) **123-2**. The provision of such an inert gas to the side wall hole **125** has the effect of preventing the oxidation of the electrode rod **123** even in a high-temperature environment when performing a necessary semiconductor process while supporting a target substrate **11** on the insulating plate **110**.

[0067] FIG. 5A is a cross-sectional view illustrating a ceramic susceptor **300** according to yet another embodiment of the present disclosure.

[0068] Referring to FIG. 5A, the ceramic susceptor **300** is a modification of the ceramic susceptor **100** of FIG. 2 and the ceramic susceptor **200** of FIG. 4A. The ceramic susceptor **300** includes a hollow shaft **120** with a groove **127**. The ceramic susceptor **300** may be configured to additionally have the groove **127** at the lowermost end surface of the hollow shaft **120** in the ceramic susceptor **100** of FIG. 2 or the ceramic susceptor **200** of FIG. 4A.

[0069] In some cases, the ceramic susceptor **300** may be configured to include the electrode rod **121** or **123** along with the hollow shaft **120** having the groove **127**, in place of the electrode rod **121** in the side wall hole **125** of the ceramic susceptor **100** of FIG. 2 or in place of the power supply conductor socket **150** and the electrode rod **123** of the susceptor **200** of FIG. 4A.

[0070] As illustrated in FIG. 5A, each electrode rod **121** or **122**, which supplies power to the electrode **112** or **114**, may be accommodated in the hollow **129** or a side wall hole **125** of the shaft **120**, may extend to the outside through the mount **140**.

[0071] Along with this, the side wall hole **125** may be provided with a rigid rod-shaped electrode rod **121**. Alternatively, the side wall hole **125** may also be provided with an electrode rod **123** having a gas flow path **123-1** and gas hole(s) **123-2**. The rigid rod-shaped electrode rod **121** and the electrode rod **123** with the gas flow path **123-1** and gas hole(s) **123-2** may coexist or only one of them may be provided.

[0072] The mount **140** includes a gas supply flow path **145** provided within the body **141**. The gas supply flow path **145** may be a through-hole formed in the rigid portion of the body **141** or a separate member mounted in the internal space in the form of a tube or pipe structure. The mount **140** may further include a gas connection portion **142** coupled to or integrated with the body **141**. The gas supply flow path **145** is in communication with the groove **127** of the hollow shaft **120**.

[0073] FIG. 5B is a view of the surface C-D of the hollow shaft **120** to which the mount **140** is coupled, as seen from below in FIG. 5A.

[0074] Referring to FIG. 5B, the groove **127** is formed on the lower end surface of the hollow shaft **120** to be recessed relative to the plane of the lower end surface. The groove **127** is in communication with the gas supply flow path **145** and may be a loop-shaped groove formed continuously in the circumferential direction. The groove **127** may extend to meet the side wall holes **125** at least once, and one or more electrode rods **121** or **123** in the side wall holes **125** may pass through the groove **127** and extend to penetrate the mount **140**.

[0075] As a result, a gas tube for supplying an inert gas (e.g., He or N.sub.2) may be connected to the gas connection portion **142**, and the gas supplied to the groove **127** through the gas supply flow path **145** in communication with the gas tube may flow along the electrode rods **121** or **123** passing through the groove **127** and may be supplied to the side wall holes **125**.

[0076] For example, when rigid electrode rods **121** (see FIG. 2) passes through the groove **127**, the gas may flow into the side wall holes **125** through the gap between the side wall hole **125** and the electrode rod **121**, thereby supplying the gas to the side wall holes **125**. Alternatively, when the electrode rods **123** each having the gas flow path **123-1** and the gas hole(s) **123-2** (see FIG. 4C)

passes through the groove **127**, the gas may flow into the side wall holes **125** through the gap between the side wall holes **125** and the electrode rods **121**. In addition, the gas may flow along the gas flow paths **123-1** within the electrode rod **123** and may be supplied to the side wall holes **125** through the gas hole(s) **123-2**.

[0077] The provision of such an inert gas to the side wall holes **125** has the effect of preventing the oxidation of the electrode rods **121** or **123** even in a high-temperature environment when performing a necessary semiconductor process while supporting a target substrate **11** on the insulating plate **110**.

[0078] Along with the oxidation prevention effect as described above, the ceramic susceptors **100**, **200**, and **300** of the present disclosure may easily prevent the risk of physical short circuits between the electrodes **121** by arranging the electrode rods **121** in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) **125** of the hollow shaft **120**, thereby eliminating the space in which the electrodes **121** shake or bend. In addition, the arrangement of the electrode rods **122** inside the hollow **129** of the hollow shaft **120** may be combined with the arrangement of the electrode rods **121** or **123** in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) **125** of the hollow shaft **120**. This combination allows for flexible design of the arrangement of electrodes **112** or **114**, depending on changes in the number or positions of the electrode rods **121**, **122**, or **123**, and enables the arrangement of a greater number of electrode rods **121**, **122**, or **123**. In addition, by arranging the electrode rods **121** or **123** in the side wall holes (e.g., $\Phi 3$ to $\Phi 6$) **125** of the hollow shaft **120**, the insulation of the hollow shaft **120** is utilized to make it unnecessary to provide separate insulating coatings for respective electrode rods **121** or **123**. In addition, by separating the radio frequency electrode rods and the heater electrode rods within the side wall holes **125** of the hollow shaft **120** or arranging them separately in the side wall holes **125** and the hollow **129**, the spacing between the rods does not narrow, thereby preventing arcing.

[0079] Furthermore, in the ceramic susceptors **100**, **200**, and **300** of the present disclosure described above, when the electrode rods **122** in the hollow **129** are used as radio frequency electrode rods and the electrode rods **121** and **123** in the side wall holes **125** are used as heater electrode rods, radio frequency (RF) noise may be prevented. For example, when performing a plasma process while supporting a target substrate **11** on the ceramic susceptor **100**, **200**, or **300**, the electrode rod **122** in the hollow **129** may be grounded, and the required power may be supplied through a radio frequency head (not illustrated) above the target substrate **11**. In this case, unlike conventional methods, since grounding is achieved solely through the electrode rod **122** in the hollow **129**, radio frequency current does not flow through the electrode rods **121** or **123** in the side wall holes **125**, which are spaced apart from the electrode rod **122** in the hollow **129** and insulated by the ceramic material of the shaft **120**. As a result, radio frequency (RF) noise may be blocked during the plasma process, thereby improving uniformity in a process such as dry etching.

[0080] In the foregoing, the present disclosure has been described based on specific details, such as concrete components, limited embodiments, and drawings, but these have been provided merely to aid a more comprehensive understanding of the present disclosure, and the present disclosure is not limited to the above-described embodiments. Various modifications and alterations may be made without departing from the essential characteristics of the present disclosure by a person ordinarily skilled in the art to which the present disclosure pertains. Therefore, the spirit of the present disclosure should not be limited to the described embodiments, and not only the appended claims, but also all technical ideas that are equivalent or have equivalent modifications to the claims should be construed as being included within the scope of the present disclosure.

Claims

1. A ceramic susceptor comprising: an insulating plate in which one or more electrodes are arranged; a hollow shaft with one end connected to the insulating plate; and one or more electrode

rods connected to the electrodes, wherein the hollow shaft comprises one or more side wall holes penetrating an interior of a side wall, wherein the one or more electrode rods comprise one or more first electrode rods, each of which has one end electrically connected to a first electrode among the one or more electrodes, the first electrode rods extending through the side wall holes, respectively, and wherein the first electrode rods each comprise a gas flow path formed of an internal empty space configured to introduce gas from another end, and one or more gas holes in communication with the gas flow path.

2. A ceramic susceptor comprising: an insulating plate in which one or more electrodes are arranged; a hollow shaft with one end connected to the insulating plate; and one or more electrode rods connected to the electrodes, wherein the hollow shaft comprises one or more convex portions that are convex radially inward from an inner surface of the side wall, the convex portions comprising side wall holes penetrating the side wall, and wherein the one or more electrode rods comprise one or more first electrode rods, each of which has one end electrically connected to a first electrode among the one or more electrodes, the one or more first electrode rods extending through the side wall holes, respectively.

3. The susceptor of claim 1, wherein the one or more electrode rods further comprise one or more second electrode rods, each of which is connected to a second electrode among the one or more electrodes, the one or more second electrode rods extending through a hollow of the hollow shaft.

4. The ceramic susceptor of claim 3, wherein the first electrodes each comprise a heater electrode, and the second electrodes each comprise a radio frequency electrode.

5. The ceramic susceptor of claim 4, wherein multiple first electrode rods are respectively connected to terminals of the heater electrodes.

6. The ceramic susceptor of claim 3, wherein the first electrodes each comprise a radio frequency electrode, and the second electrodes each comprise a heater electrode.

7. The ceramic susceptor of claim 6, wherein multiple second electrode rods are respectively connected to terminals of the heater electrodes.

8. The ceramic susceptor of claim 1, further comprising: a conductive socket coupled to the electrode rods and configured to supply power, wherein the conductive socket comprises a gas supply flow path configured to communicate with the gas flow path.

9. The ceramic susceptor of claim 1, further comprising: a mount coupled to a lower portion of the hollow shaft.

10. The ceramic susceptor of claim 9, wherein the hollow shaft comprises a groove formed on a surface coupled to the mount and extending to meet the side wall holes at least once, wherein the mount comprises a gas supply flow path configured to communicate with the groove, and wherein the one or more first electrode rods pass through the groove.

11. The ceramic susceptor of claim 9, wherein the one or more electrode rods further comprise one or more second electrode rods, each of which is connected to a second electrode among the one or more electrodes, the one or more electrode rods extending through a hollow of the hollow shaft, and wherein the one or more second electrode rods extend through the mount.
